

# Package ‘manMetaVAR’

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Coefficients

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coef.manmetavar.dtvvar *Parameter Estimates (FitDTVAR)*

---

### Description

Parameter Estimates (FitDTVAR)  
 Parameter Estimates (FitMetaVAR)  
 Parameter Estimates (FitMplus)

### Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
```

```

    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, ...)

```

### Arguments

|                 |                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object          | Object of class <code>manmetavar.mplus</code> .                                                                                                                     |
| alpha           | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| beta            | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| nu              | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| psi             | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| theta           | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                                              |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |
| vanishing_theta | Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .                                                                       |
| theta_tol       | Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .                                            |
| ...             | additional arguments.                                                                                                                                               |
| median          | Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.                             |

### Value

Returns a list of vectors of parameter estimates.

Returns a vector of estimated parameters.

Returns a vector of estimated parameters.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Compress

*Compress Replication*

---

**Description**

Compress Replication

**Usage**

```
Compress(taskid, repid, output_folder)
```

**Arguments**

|               |                                   |
|---------------|-----------------------------------|
| taskid        | Positive integer. Task ID.        |
| repid         | Positive integer. Replication ID. |
| output_folder | Character string. Output folder.  |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

confint.manmetavar.metavar

*Confidence Intervals for the Parameter Estimates (FitMetaVAR)*

---

**Description**

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Confidence Intervals for the Parameter Estimates (FitMplus)

**Usage**

```
## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, ...)
```

Arguments

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.mplus</code> .                                                                                                                       |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| lb     | Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.                                                 |
| ...    | additional arguments.                                                                                                                                                 |

Value

Returns a matrix of confidence intervals.  
Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

---

|          |                         |
|----------|-------------------------|
| Dynamics | <i>Process Dynamics</i> |
|----------|-------------------------|

---

Description

Process Dynamics

Usage

`Dynamics(dynamics)`

Arguments

|          |                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------|
| dynamics | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |
|----------|--------------------------------------------------------------------------------------------------------------|

See Also

Other Data Generation Functions: [GenData\(\)](#)

**Examples**

```
## Not run:
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)

## End(Not run)
```

FitDTVAR

*Fit the Model using the fitDTVARMxID Package***Description**

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

**Usage**

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |
| seed   | Integer. Random seed.                                       |

**See Also**

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

**Description**

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

**Usage**

```
FitMetaVAR(fit, ncores = NULL)
```

**Arguments**

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| <code>fit</code>    | R object. Output of the <a href="#">FitDTVAR()</a> function. |
| <code>ncores</code> | Positive integer. Number of cores to use.                    |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

---

FitMLVAR

*Fit the Model using the mlVAR Package*


---

### Description

The function fits the model using the [mlVAR::mlVAR](#) package.

### Usage

```
FitMLVAR(data, ncores = NULL)
```

### Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |

### See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

### Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)

## End(Not run)
```

---

FitMplus

*Fit the Model using Mplus*


---

### Description

The function fits the model using Mplus.

### Usage

```
FitMplus(data, wd = ".", mplus_bin = NULL, ncores = NULL)
```



**Arguments**

|           |                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------|
| data      | R object. Output of the <a href="#">GenData()</a> function.                                                        |
| wd        | Character string. Working directory.                                                                               |
| mplus_bin | Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary. |
| ncores    | Positive integer. Number of cores to use.                                                                          |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

---

GenData

*Simulate Data*

---

**Description**

The function simulates data using the [simStateSpace::SimSSMIVary\(\)](#) function.

**Usage**

```
GenData(taskid)
```

**Arguments**

|        |                            |
|--------|----------------------------|
| taskid | Positive integer. Task ID. |
|--------|----------------------------|

**See Also**

Other Data Generation Functions: [Dynamics\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)

## End(Not run)
```

---

MplusInput

---

Create Mplus Input File

---

**Description**

The function creates an Mplus input file.

**Usage**

```
MplusInput(
  fn_data = NULL,
  fn_estimates = NULL,
  fn_results = NULL,
  fn_posterior = NULL,
  fn_factorscores = NULL,
  ncores = NULL
)
```

**Arguments**

|                 |                                                      |
|-----------------|------------------------------------------------------|
| fn_data         | Character string. Filename for data file.            |
| fn_estimates    | Character string. Filename for estimates output.     |
| fn_results      | Character string. Filename for results output.       |
| fn_posterior    | Character string. Filename for posterior output.     |
| fn_factorscores | Character string. Filename for factor scores output. |
| ncores          | Positive integer. Number of cores to use.            |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

**Examples**

```
cat(MplusInput())
```

---

|        |                              |
|--------|------------------------------|
| params | <i>Simulation Parameters</i> |
|--------|------------------------------|

---

**Description**

Simulation Parameters

**Usage**

params

**Format**

A dataframe with 25 rows and 3 columns:

**taskid** Simulation Task ID.**n** Sample size.**time** Number of measurement occasions.**dynamics** Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| plot.manmetavar.data | <i>Plot Method for an Object of Class manmetavar.data</i> |
|----------------------|-----------------------------------------------------------|

---

**Description**

Plot Method for an Object of Class manmetavar.data

**Usage**

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

**Arguments**

|     |                                  |
|-----|----------------------------------|
| x   | Object of class manmetavar.data. |
| ... | additional arguments.            |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

print.manmetavar.data *Print Method for an Object of Class manmetavar.data*

---

## Description

Print Method for an Object of Class manmetavar.data

Print Method (FitDTVAR)

Print Method (FitMetaVAR)

Print Method (FitMLVAR)

## Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.dtvvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

**Arguments**

|                 |                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x               | Object of class <code>manmetavar.mlvar</code> .                                                                                                                     |
| ...             | additional arguments.                                                                                                                                               |
| means           | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                                |
| alpha           | Numeric vector. Significance level $\alpha$ .                                                                                                                       |
| beta            | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| nu              | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| psi             | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| theta           | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                                              |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |
| vanishing_theta | Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .                                                                       |
| theta_tol       | Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .                                            |
| digits          | Integer indicating the number of decimal places to display.                                                                                                         |
| show            | Tables to show.                                                                                                                                                     |

**Value**

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |                               |
|-----|-------------------------------|
| Sim | <i>Simulation Replication</i> |
|-----|-------------------------------|

---

**Description**

Simulation Replication

**Usage**

```
Sim(taskid, repid, output_folder, overwrite, integrity, seed, mplus = FALSE)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| repid         | Positive integer. Replication ID.                                                         |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed          | Integer. Random seed.                                                                     |
| mplus         | Logical. If mplus = TRUE, fit a DSEM model in Mplus.                                      |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                                          |
|-------------|------------------------------------------|
| SimFitDTVAR | <i>Simulation Replication - FitDTVAR</i> |
|-------------|------------------------------------------|

---

**Description**

Simulation Replication - FitDTVAR

**Usage**

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitMetaVAR

*Simulation Replication - FitMetaVAR*

---

**Description**

Simulation Replication - FitMetaVAR

**Usage**

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                                          |
|-------------|------------------------------------------|
| SimFitMLVAR | <i>Simulation Replication - FitMLVAR</i> |
|-------------|------------------------------------------|

---

**Description**

Simulation Replication - FitMLVAR

**Usage**

SimFitMLVAR(taskid, repid, output\_folder, seed, suffix, overwrite, integrity)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| repid         | Positive integer. Replication ID.                                                         |
| output_folder | Character string. Output folder.                                                          |
| seed          | Integer. Random seed.                                                                     |
| suffix        | Character string. Output of manCTMed::SimSuffix().                                        |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan



---

|             |                                          |
|-------------|------------------------------------------|
| SimFitMplus | <i>Simulation Replication - FitMplus</i> |
|-------------|------------------------------------------|

---

**Description**

Simulation Replication - FitMplus

**Usage**

```
SimFitMplus(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|       |                             |
|-------|-----------------------------|
| SimFN | <i>Simulation File Name</i> |
|-------|-----------------------------|

---

**Description**

Simulation File Name

**Usage**

```
SimFN(output_type, output_folder, suffix)
```

**Arguments**

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| output_type   | Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var". |
| output_folder | Character string. Output folder.                                                                                  |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> .                                                  |

**Value**

Returns a character string file name with the `output_folder` in the OS-specific format.

---

|            |                                         |
|------------|-----------------------------------------|
| SimGenData | <i>Simulation Replication - GenData</i> |
|------------|-----------------------------------------|

---

**Description**

Simulation Replication - GenData

**Usage**

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> .                                                      |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|         |                                |
|---------|--------------------------------|
| SimProj | <i>Simulation Project Name</i> |
|---------|--------------------------------|

---

**Description**

Simulation Project Name

**Usage**

SimProj()

**Value**

Returns the project name as a character string.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |                |
|-----|----------------|
| Sum | <i>Summary</i> |
|-----|----------------|

---

**Description**

Summary

**Usage**

Sum(taskid, reps, output\_folder, overwrite, integrity, mplus = FALSE)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| mplus         | Logical. If mplus = TRUE, fit a DSEM model in Mplus.                                      |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SumFitMetaVARLB

*Summary: Likelihood-Based (FitMetaVAR)*


---

**Description**

Summary: Likelihood-Based (FitMetaVAR)

**Usage**

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SumFitMetaVARNormal      *Summary: Normal Theory (FitMetaVAR)*

---

**Description**

Summary: Normal Theory (FitMetaVAR)

**Usage**

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                           |
|-------------|---------------------------|
| SumFitMLVAR | <i>Summary (FitMLVAR)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMLVAR)

**Usage**

SumFitMLVAR(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

- taskid            Positive integer. Task ID.
- reps             Positive integer. Number of replications.
- output\_folder   Character string. Output folder.
- overwrite       Logical. Overwrite existing output in output\_folder.
- integrity       Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                           |
|-------------|---------------------------|
| SumFitMplus | <i>Summary (FitMplus)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMplus)

**Usage**

SumFitMplus(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

summary.manmetavar.data

*Summary Method for an Object of Class manmetavar.data*

---

**Description**

Summary Method for an Object of Class manmetavar.data

Summary Method (FitDTVAR)

Summary Method (FitMetaVAR)

Summary Method (FitMLVAR)

Summary Method (FitMplus)

**Usage**

```
## S3 method for class 'manmetavar.data'
summary(object, ...)
```

```
## S3 method for class 'manmetavar.dtvvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
```

```

    psi = TRUE,
    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    digits = 4,
    ...
)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, ...)

```

## Arguments

|           |                                                                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object    | Object of class <code>manmetavar.mplus</code> .                                                                                                                     |
| ...       | additional arguments.                                                                                                                                               |
| means     | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                                |
| alpha     | Numeric vector. Significance level $\alpha$ .                                                                                                                       |
| beta      | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| nu        | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| psi       | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| theta     | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| converged | Logical. Only include converged cases.                                                                                                                              |
| grad_tol  | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| hess_tol  | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |



|                 |                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                             |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                            |
| digits          | Integer indicating the number of decimal places to display.                                                 |
| show            | Tables to show.                                                                                             |
| median          | Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior. |

### Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, number of posterior samples, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

vcov.manmetavar.dtvvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

---

### Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMplus)

### Usage

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, ...)

```

### Arguments

|                              |                                                                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>          | Object of class <code>manmetavar.mplus</code> .                                                                                                                     |
| <code>alpha</code>           | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| <code>beta</code>            | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| <code>nu</code>              | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| <code>psi</code>             | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| <code>theta</code>           | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| <code>converged</code>       | Logical. Only include converged cases.                                                                                                                              |
| <code>grad_tol</code>        | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| <code>hess_tol</code>        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |
| <code>vanishing_theta</code> | Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .                                                                       |
| <code>theta_tol</code>       | Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .                                            |
| <code>...</code>             | additional arguments.                                                                                                                                               |

### Value

Returns a list of sampling variance-covariance matrices.

Returns the sampling variance-covariance matrix of the estimated parameters.

Returns the sampling variance-covariance matrix of the estimated parameters.

### Author(s)

Ivan Jacob Agaloos Pesigan

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